

JAEHAN IM

Austin, Texas, USA [◇ \[LinkedIn\]](#) [◇ \[Scholar\]](#) [◇ \[Webpage\]](#)
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EDUCATION

The University of Texas at Austin

Ph.D. candidate in Department of Aerospace Engineering and Engineering Mechanics
◇ Advisor: Dr. Ufuk Topcu, Dr. David Fridovich-Keil

Aug. 2023 - present
Austin, Texas, USA

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in the Department of Aerospace Engineering

◇ Advisor: Jaemyung Ahn, Ph.D.

◇ Thesis: “Distributed free-flow air traffic management system based on Nash equilibrium concept”

Feb. 2020
Daejeon, Republic of Korea

Delft University of Technology

Exchange student of the Department of Aerospace Engineering

Feb. 2017
Delft, Netherlands

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in the Department of Aerospace Engineering

◇ Minor in the Department of Civil and Environmental Engineering

◇ Magma Cum Laude

Feb. 2018
Daejeon, Republic of Korea

EXPERIENCES

Ecole Nationale de l’Aviation Civile, Visiting Scholar, Optimization Team (Prof. Daniel Delahaye)
Toulouse, France

Feb. 2025

- ◇ Formulated airspace sector overload mitigation as a noncooperative game governed by best-response dynamics.
- ◇ Analyzed convergence and stability properties under potential-game structures.
- ◇ Designed decentralized coordination mechanisms for congestion resolution with provable feasibility guarantees.

Nearthlab Inc., Research Engineer, Flight Control Department
Seoul, Republic of Korea

Feb. 2020 – May 2023

- ◇ Developed autonomous flight guidance systems for infrastructure inspection.
- ◇ Designed quantitative evaluation methods for aerial vehicle performance.
- ◇ Researched scalable optimal multi-drone routing algorithms for cooperative and industry-scale missions.

CnTech Inc., System Engineer
Daejeon, Republic of Korea

Apr. 2017 – Apr. 2018

- ◇ Contributed to concept-of-operations and PDR simulations for stand-alone IoT generator systems.

SELECTED PUBLICATIONS

Journal

- [1] **Scaling Chance-Constrained Correlated Equilibrium Computation for Exclusive Resource-Assignment Games**
J. Im, U. Topcu, D. Fridovich-Keil
IEEE Control Systems Letters (L-CSS). (Under Review) [\[Link\]](#)
- [2] **Chance-Constrained Correlated Equilibrium for Robust Noncooperative Coordination**
J. Im, U. Topcu, D. Fridovich-Keil
IEEE Control Systems Letters (L-CSS). (Accepted) [\[Link\]](#)
- [3] **Iterative Negotiation and Oversight: A Case Study in Decentralized Air Traffic Management**
J. Im, J. Clarke, U. Topcu, D. Fridovich-Keil
Elsevier Transportation Research Part C. (Accepted) [\[Link\]](#)
- [4] **Game-theoretic Regulated Decentralized Coordination for Airspace Sector Overload Mitigation**
J. Im, D. Delahaye, D. Fridovich-Keil, U. Topcu
Elsevier Transportation Research Part B. (Under review) [\[Link\]](#)

- [5] **Noncooperative Coordination via a Trading-based Auction**
J. Im, F. Fotiadis, D. Delahaye, U. Topcu, D. Fridovich-Keil
IEEE Transactions on Automatic Control (TAC). (Published) [\[Link\]](#)
- [6] **Coordination in Noncooperative Multiplayer Matrix Games via Reduced Rank Correlated Equilibria**
J. Im, Y. Yu, D. Fridovich-Keil, U. Topcu
IEEE Control Systems Letters (L-CSS). (Published) [\[Link\]](#)
- [7] **Decentralized Free-Flow Traffic Management Based on Nash Equilibrium**
J. Im, J. Ahn
AIAA Journal of Aerospace Information System (JAIS), 2022. (Published) [\[Link\]](#)
- [8] **Multi-Agent Inspection Path Planning with Large-Scale Vehicle Routing Problem**
J. Im, B.Y. Lee
AIAA Journal of Aerospace Information System (JAIS), 2023. (Published) [\[Link\]](#)
- [9] **Modeling and simulation of aircraft motion for performance assessment of airborne AESA radar considering wind and vibration**
D. Lee, J. Im, H. Lee, Y. Jung, J. Jeong, J. Shin, S. Lee, J. Park, J. Ahn
Journal of the Korean Society for Aeronautical & Space Sciences (JKSAS), 2020. (Published) [\[Link\]](#)

Conference

- [1] **Secure Coordination for Vertiport Sequencing in Advanced Air Mobility**
J. Im, F. Fotiadis, U. Topcu, D. Fridovich-Keil
2027 AIAA SciTech Forum, 2027 (Accepted) [\[Link\]](#)
- [2] **Noncooperative Coordination for Decentralized Air Traffic Management**
J. Im
US-Europe Air Transportation Research & Development Symposium 2026, 2026 (Accepted) [\[Link\]](#)
- [3] **Chance-Constrained Correlated Equilibrium for Robust Noncooperative Coordination**
J. Im, U. Topcu, D. Fridovich-Keil
2026 IEEE Conference on Decision and Control (CDC), 2026. (Accepted) [\[Link\]](#)
- [4] **Game-theoretic Coordination of Decentralized Air Traffic Management**
J. Im, F. Fotiadis, J. Clarke, D. Delahaye, D. Fridovich-Keil, U. Topcu,
2025 SESAR Innovation Days Conference, 2025. (Poster accepted) [\[Link\]](#)
- [5] **Coordination in Noncooperative Multiplayer Matrix Games via Reduced Rank Correlated Equilibria**
J. Im, Y. Yu, D. Fridovich-Keil, U. Topcu
2024 63rd IEEE Conference on Decision and Control (CDC), 2024. (Published) [\[Link\]](#)
- [6] **Spatiotemporal VRP for Collision-free Multi-UAV Inspection Planning**
J. Im, Y. Kim
IEEE International Conference on Unmanned Aircraft Systems (ICUAS), 2023. (Published) [\[Link\]](#)
- [7] **Collision-free Vehicle Routing Problem for multiple drone infrastructure inspection**
J. Im
In Proceedings of the Korean Society for Aeronautical & Space Sciences fall conference, 2022.
- [8] **VRP based 3D optimal path planning for industrial facility inspection**
J. Im, J. Lee
In Proceedings of the Korean Society for Aeronautical & Space Sciences spring conference, 2022.
- [9] **Optimal path planning for spatiotemporal space with grid search algorithm**
J. Im, J. Ahn
In Proceedings of the Korean Society for Aeronautical & Space Sciences fall conference, 2019.
- [10] **Wind and fuselage vibration considered 6DOF flight simulator development for AESA radar testing**
J. Im, D. Lee, H. Lee, H. Lee, Y. Jung, J. Jeong, H. Park, J. Ahn
In Proceedings of the Korean Society for Aeronautical & Space Sciences spring conference, 2019.
- [11] **Communication-less decentralized ATC with Nash equilibrium strategy and individual observance-based belief system**
J. Im, J. Ahn
In Proceedings of the Korean Society for Aeronautical & Space Sciences fall conference, 2018.

Books and Book Chapters

- [1] **Game-Theoretic Coordination of Noncooperative Agents for Safe Urban Air Mobility Operations**
J. Im, D. Fridovich-Keil, U. Topcu
In Handbook of Urban Air Mobility and Drone Integration, Springer. (Book chapter, Under review)
- [2] **Flying**
J. Im
Across Books, ISBN: 979-11-6774-113-4, 2023.
Recipient of the 2023 Excellent Publication Content Award from the Korea Publication Industry Promotion Agency.

RESEARCH PROJECTS

- [1] **Scalable and Uncertainty-Aware Correlated Equilibrium**, 2023 – present
 - ◊ Developed scalable algorithms for computing correlated equilibria in large joint-action spaces using reduced-rank and convex-analytic formulations.
 - ◊ Formulated uncertainty-aware and chance-constrained equilibrium models that preserve incentive compatibility under noisy or incomplete payoff information.
 - ◊ Established convergence, feasibility, and efficiency guarantees, validated through large-scale simulations in decentralized safety-critical coordination settings.
- [2] **Mechanism Design and Oversight in Noncooperative Coordination**, 2024 – present
 - ◊ Developed auction-based mechanisms for decentralized equilibrium selection among self-interested agents with private information.
 - ◊ Introduced an oversight framework that steers collective outcomes without centralized enforcement or direct pairwise negotiation.
 - ◊ Established finite-time convergence and incentive-compatibility guarantees under constrained and safety-critical settings.
- [3] **Dynamic Incentive Control for Congestion and Overload Mitigation**, 2025 – present
 - ◊ Modeled congestion and resource overload as strategic multi-agent games governed by best-response dynamics.
 - ◊ Designed dynamic incentive policies that reshape equilibrium structure while preserving decentralized decision authority.
 - ◊ Analyzed stability and convergence properties using potential and ordinal potential frameworks with provable recovery guarantees.
- [4] **Human–Autonomy Interaction and Intention Inference**, 2024 – 2025
 - ◊ Developed game-theoretic models to infer human cognitive states in shared-control systems.
 - ◊ Formulated and solved inverse game problems to detect attention shifts and mode confusion.
 - ◊ Designed adaptive autonomy mechanisms that reallocate control authority based on inferred intent.
- [5] **Industry-scale optimal routing algorithms for multi-drone systems**, 2021 – 2022
 - ◊ Developed scalable routing algorithms for cooperative multi-UAV infrastructure inspection missions.
 - ◊ Incorporated collision-free constraints and graph-clustering techniques to handle densely located nodes.
 - ◊ Developed Louvain algorithm-based method to accelerate solution time for large problem instances (1k–2k nodes).
- [6] **6DOF flight simulation algorithm development for radar performance simulation**, 2018 – 2019
 - ◊ Developed a simulator that mimics high-frequency vibration in flight.
 - ◊ Structural vibration reconstructed with actual PSD data in flight.
 - ◊ Developed using MATLAB Simulink.

TECHNICAL SKILLS

Programming: Python (PyTorch), MATLAB/Simulink, Julia, C++, CVX, Gurobi, Git, Linux, L^AT_EX

HONORS AND FELLOWSHIPS

- [1] **Cockrell School of Engineering Fellowship** 2023 - present
- [2] **Professional Development Award** Jun. 2026
 - ◊ Awarded by The University of Texas at Austin Graduate School
- [3] **2023 Excellent Publication Content Award** Jun. 2023
 - ◊ Awarded by the Korea Publication Industry Promotion Agency for the book *Flying*

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| <p>[4] Top award (1st place) at 2019 Korea Aerospace Paper Awards</p> <ul style="list-style-type: none"> ◊ Won the top prize (Minister of Trade, Industry and Energy's award). ◊ Title: "Distributed free-flow traffic management system based on Nash equilibrium concept". <p>[5] Undergraduate Scholarship from Boeing Korea</p> | <p>Oct. 2019</p> <p>2014 - 2016</p> |
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TEACHING EXPERIENCE

- ◊ Teaching assistant, Linear Systems Analysis (ASE 330M), UT Austin, 2026 spring.
- ◊ Undergraduate student mentoring (Seth L. Cohen, UT Austin), 2026 spring.
- ◊ NASA ULI summer intern mentoring (Jocelyn Pelegrino, Morgan State University), 2025 summer.
- ◊ Teaching assistant, Linear Systems Analysis (ASE 330M), UT Austin, 2025 spring.
- ◊ Teaching assistant, Feedback Control Systems (ASE 370C), UT Austin, 2025 spring.
- ◊ Undergraduate student mentoring (Nir Miller, UT Austin), 2024 Fall.

INVITED ACADEMIC TALKS

- ◊ <Application of Game Theory in ATM Systems>, Ecole Nationale de l'Aviation Civile (ENAC), 2025.
- ◊ <Noncooperative Coordination in Aerospace Systems>, Korean-American Sci. and Eng. Association (KSEA), 2025.
- ◊ <Decentralized and Noncooperative Coordination>, Center for Autonomy Seminar, 2024.

PROFESSIONAL ACTIVITIES AND PUBLIC ENGAGEMENT

- ◊ Featured speaker, <The Use of Time in Aviation>, Cartier Le Salon de Santos, 2025.
- ◊ Featured speaker, <Jaehan's business soaring through the skies>, EBS Business Review, 2025.
- ◊ Featured guest, <Gwedo x Jaehan>, Space Hub TV, Hosted by science communicator Gwedo, 2024. [YouTube]

PATENTS

- [1] **Jaehan Im**, Byung Yoon Lee, "Method and Apparatus for Generating Movement Paths of Plurality of Drones for Inspection of Industrial Structure," U.S. Patent Application No. 18/868,109, filed Nov. 21, 2024.
- [2] **Jaehan Im**, Byung Yoon Lee, "Method and Apparatus for Generating Movement Paths of Plurality of Drones for Inspection of Industrial Structure," PCT International Application No. PCT/KR2023/007669, filed Jun. 5, 2023.
- [3] **Jaehan Im**, Byung Yoon Lee, "Method and Apparatus for Generating a Drone Movement Path for Wind Turbine Inspection," Korean Patent No. 10-2431247, registered Aug. 5, 2022.
- [4] **Jaehan Im**, Byung Yoon Lee, "Method and Apparatus for Generating a Drone Movement Path for Inspection of Industrial Structures," Korean Patent No. 10-2431251, registered Aug. 5, 2022.
- [5] **Jaehan Im**, Byung Yoon Lee, "Method and Apparatus for Generating Movement Paths of Multiple Drones for Inspection of Industrial Structures," Korean Patent No. 10-2431252, registered Aug. 5, 2022.
- [6] Minwoo Koh, Kiduk Kim, **Jaehan Im**, Kangmin Sun, Jeonghun Lee, "Mold Asset Management System," Korean Patent No. 10-2205304, registered Jan. 14, 2021.